

A Multicriteria Trading System Based on Singular Spectrum Analysis Trading Rules

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Abstract—A new trading system (TS) is proposed in this paper, based on singular spectrum analysis (SSA) and on a decision-making process that relies on performance measures adjusted by profit and risk. The aim is to contribute to the literature of technical analysis (TA) by proposing a general and adaptive tool for supporting trading strategies of investors and market practitioners. Recent advances of trading rules based on SSA are pooled together in a complimentary fashion, with the intention that this group of strategies outperforms its individual constituents, as occur with ensembles. Simulations are presented showing the improvements of the proposed TS over classic tools of TA and the benchmark Buy & Hold strategy for the Brazilian stock market.

Index Terms—Multiple-criteria decision-making, singular spectrum analysis (SSA), technical trading rules (TTR), trading system (TS).

I. INTRODUCTION

STOCK markets the meeting points for buyers and sellers of securities, are regarded as very complex financial systems in which companies can gather financing, and investors become shareholders, profiting from its future performance. It has been understood that stock markets are perfectly efficient systems, which process information in an unbiased manner, the enduring efficient market hypothesis (EMH) [1]. Nevertheless, many statistical patterns have been reported confronting this understanding: long-run reversals, seasonal patterns (day-of-the-week, holidays, “the January effect”), speculative bubbles, and financial crisis [2], [3].

In fact, stock markets are nonlinear and nonstationary systems, influenced by a bundle of interrelated factors, outside and inside the realm of economics. Despite challenges and controversial aspects, everyday practice reveals that many investors seek to hedge risk [4] or achieve above-average earnings. This statistical procedure, in which an investor seeks generalizations

in historical price series that can be exploited in the future, is known as technical analysis (TA). A compelling theory used to justify the predictive capacity of TA is the adaptive market hypothesis (AMH), which is presented in [5] as an alternative that reconciles EMH with behavioral economics. An in-depth discussion regarding AMH is available in [6]–[8].

In this context, trading systems (TS) are used to support investments, with a considerable body of authors advocating in its favor over the years, (see [9] and references therein). An international survey conducted in five countries reported that 86% of fund managers rely their investment management on TS [10]. In practice, the development of new computational techniques and increasing availability of data fostered the expansion of TS, including high frequency trading. This added even more complexity to the scenario [11], [12].

Briefly, TS aggregate some input signals and, based on a set of parameters, create buy/sell recommendations of a given security as outputs. Many input signals have been used, including: observation of stocks time-series and diverse indexes [13]; fundamental data (revenue, cash flow, etc); behavioral metrics (messages in social network services, sentiment analysis [14], etc.); or even simulated models [15]. In general, these information are referred simply as real-world data. The outputs can be sent to a stockbroker, in order to support his/her trade decision, or can be executed by software, automatically.

This paper proposes a new TS based on recent developments in the field of computational finance. A system of systems (SoS) approach is adopted, which intends to produce an adaptive trading process that evolves over time, serving as a support to the decision-making of an investor or market practitioner. Stock time-series are the real-world data for the proposed TS, which are processed using the singular spectrum analysis (SSA), serving as technical indicators (TIs).

SSA is a nonparametric technique that is adaptive to the underlying data, able to discriminate meaningful components in general time-series, being applied in financial problems [16]. Recently, in [17]–[20], technical trading rules (TTRs) were designed based on SSA, with similar or superior performance than a wide range of classic strategies from TA. One of the advantages of using SSA for TS is the possibility to analyze financial data at various time scales using a single framework.

Instead of committing to a single trading strategy over time, the proposed TS pools several SSA-TTRs, choosing the more adequate strategy, in given a time-frame. The idea is that the combination of subsystems offers complimentary information in a

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scenario of uncertainty, as occurs in stock markets, and they can perform better together than alone. Performance metrics, which are not only based on profitability but also on risk-adjustment, are used to evaluate the subsystems and support its selection according to the market conditions. Therefore, the proposed TS is flexible enough to allow the operator to select the best approach, given a performance metric of his/her choice, or to establish a multiple-criteria decision-making (MCDM), taking into account several metrics simultaneously.

In the light of the AMH, profitability of TS can coexist with the EMH, since the success of investment strategies may vary over time, according to the market conditions. An ecology of investors, competing amongst themselves, creating, discarding, and adopting several trading strategies results in an adaptation process that have many timescales [21]. Therefore, the adaptive and hybrid characteristics of the proposed TS may yield an improved performance, since it also works on multiple timescales because of the presence of SSA and MCDM.

II. RELATED WORKS

Majority of TS uses solely the time-series of the security as input. The raw time-series is processed, generating a TI, aiming to unveil patterns of interest. Based on the manipulated data, buying/selling decisions are taken, according to the so-called TTRs.

On a historical note, candlestick patterns are the earliest usage of TS, tracing back to 1700s, in rice markets in Japan [22]. Works of Charles Dow and William P. Hamilton, in the turn of the 20th century within Wall Street Journal, are also notorious. However, the first scientific paper ever published with TS is regarded as [23].

Market movements are considered fundamental in TA. Trends and oscillations, popular features in stock markets, are exploited to assess prices in long- and short-time periods. A cornerstone paper in TA is the work [24], in which simple techniques were backtested in the Dow Jones industrial average (DJIA) from 1897 to 1986, revealing positive returns. Since then, many variations based on the trend-following and oscillator TIs have been developed, tested, and studied [9]. Advantages of such classical techniques are the good interpretability and the popularity among market practitioners.

The increase of computational power and data availability fostered the development of TS based on machine learning (ML). Neural networks (NNs) are employed in [25] to forecast day-ahead prices. An adaptive NN is employed by [26], which uses particle swarm algorithm for training the TS. Comprehensive reviews can be found in [27], [28]. Nevertheless, a criticism on ML in financial systems is the low interpretability [29]. Also, the aforesaid works intend price forecast, a much harder task than to detect its direction, e.g., bearish or bullish tendencies, which can be sufficient to obtain profits or hedge risks with classical TS.

Recently, another trend has arisen on the literature. This consists in efforts to capture behavioral metrics from online social networks and general media outlets to improve TS. For instance, in [14], a web crawler is used to gather relevant texts from Internet. Then, a daily sentiment index is produced based on sentence-based analysis of collected data. This index is modified by taking into account the day-of-the-week effect. Finally, a support vector machine is used to forecast the stock market movement.

Since stock markets are nonlinear and nonstationary systems, they have been attracting the attention of academics from finance and other branches of knowledge. Econophysics urged as a result of these efforts, a multidisciplinary field that relies on statistical and probabilistic methods to solve economic questions [30]. One of such statistical tools is the SSA, which has been adapted for stock markets recently in [17]–[20], serving to design TTRs. SSA has few parameters to be adjusted, thus it does not suffer from the curse of dimensionality, recurrent in other TS with very high dimensional input and parameter space. Another interesting feature is the possibility to associate its parameters with short- and long-time market movements in an understandable way for market practitioners.

III. BACKGROUND

This section briefly presents SSA and lays down some concepts of computational finance.

A. Singular Spectrum Analysis

An in-depth discussion concerning SSA is given in [31]. Briefly, SSA has two complimentary stages, each containing two steps. Let $\mathbf{x} = [x_0, x_1, \dots, x_{N-1}]^T$ be a time-series of length N and suppose that it can be approximated by a sum of distinct components $\mathbf{x} \cong \sum_r \tilde{\mathbf{x}}_r$. In the first, defining L as the embedding dimension, SSA maps the original time-series into K length L lagged vectors $\mathbf{x}_k$, where each one is given by $\mathbf{x}_k = [x_{k-L+1}, \dots, x_{k-1}, x_k]^T$, with $L \leq K$ and $K = N - L + 1$. Then, the vectors are arranged in a $L \times K$ Hankel matrix, called trajectory matrix $\mathbf{A} = [\mathbf{x}_{L-1}, \mathbf{x}_L, \dots, \mathbf{x}_{N-1}]$. In the second step, $\mathbf{A}$ is factorized by means of singular-value decomposition, resulting in R elementary matrices

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T = \sum_{r=0}^{R-1} \sigma_r \mathbf{u}_r \mathbf{v}_r^T = \sum_{r=0}^{R-1} \mathbf{A}_r \quad (1)$$

where $R = \text{rank}(\mathbf{A}) \leq L \leq K$; matrix $\mathbf{U} = [\mathbf{u}_0, \dots, \mathbf{u}_{R-1}]$, of dimension $L \times L$, and matrix $\mathbf{V} = [\mathbf{v}_0, \dots, \mathbf{v}_{R-1}]$, of dimension $K \times K$ are unitary matrices; $\mathbf{\Sigma}$ is a diagonal matrix, with dimension of $L \times K$, whose diagonal elements $\sigma_r = \sqrt{\lambda_r}$ are the singular values of $\mathbf{A}$; λ_r and $\mathbf{u}_r = [u_0^{(r)}, \dots, u_{L-1}^{(r)}]^T$ are the respective eigenvalues and eigenvectors of $\mathbf{A}\mathbf{A}^T$; and $\mathbf{A}_r$ are the elementary matrices. Moving on to the second stage, matrices $\mathbf{A}_r$ can be clustered in disjoint subsets, leading to the resultant matrix $\mathbf{A}^{\{I_m\}} = \sum_{i \in I_m} \mathbf{A}_i$, where $I_m = \{i_0, i_1, \dots, i_{p-1}\} \subset \{0, 1, \dots, R-1\}$. In the final step, for a given grouping I_m , a diagonal averaging maps the resultant matrix to a reconstructed time-series: $\langle \mathbf{A}^{\{I_m\}} \rangle \mapsto \tilde{\mathbf{x}}^{\{I_m\}}$, where $\langle \bullet \rangle$ means the averaging along the antidiagonals applied to the argument.

B. Online Data Processing With SSA

In online applications, such as TS, SSA algorithm must be recalculated for each new data x_n arriving in the growing time-series $\mathbf{x}$. The augmented time-series implies in a new lagged vector $\mathbf{x}_n$, an augmented trajectory matrix, and updated eigenvectors $\mathbf{u}_r$ and eigenvalues λ_r . Thus, an augmented reconstructed time-series $\tilde{\mathbf{x}}^{\{I_m\}}$ of length $n+1$ is generated.

TABLE I
PROFIT PERFORMANCE METRICS

Measure	Formulation
Profit Factor	$F = \frac{\sum_{p=1}^P S_p^+}{\sum_{q=1}^Q S_q^-}$
Payoff Ratio	$\mathcal{P} = \overline{P^+} / \overline{P^-}$
Expected Value	$E = \left[\left(1 + \overline{P^+} / \overline{P^-} \right) P_r^+ \right] - 1$

However, to avoid cumbersome computations, the last value of reconstructed time-series $\tilde{x}_n^{\{I_m\}}$ can be obtained directly from updated eigenvectors $\mathbf{u}^{\{I_m\}}$

$$\tilde{x}_n^{\{I_m\}} = \sum_{l=0}^{L-1} \left(\sum_{i \in I_m} u_{L-1}^{\{i\}} u_l^{\{i\}} \right) x_{n-L+l+1}. \quad (2)$$

In this paper, this procedure will be nominated “*Causal SSA*” [17], borrowing the definition of causality from the signal processing field. Alternatively, the “*Recursive SSA*” was proposed in [32], [33], consisting of changes in the computation of the trajectory matrix. To emphasize the latest values, [32] proposes the use of a weighted trajectory matrix

$$\mathbf{C} = (\text{tr}(\Phi))^{-1} \mathbf{A} \Phi \mathbf{A}^T \quad (3)$$

where Φ is a diagonal matrix composed by exponential weights $\{\varphi^{K-1}, \varphi^{K-2}, \dots, \varphi, 1\}$, such that $0 \ll \varphi < 1$, and φ is called forgetting factor. At each new value x_n , the trajectory matrix must be computed, since it is augmented with the new lagged vector $\mathbf{x}_n = [x_{n-L+1}, \dots, x_n]^T$. Therefore, a new trajectory matrix is calculated

$$\mathbf{C}_n = \varphi \mathbf{C}_{n-1} + (1 - \varphi) \mathbf{x}_n \mathbf{x}_n^T. \quad (4)$$

The eigenvectors $\mathbf{c}_i$ computed from matrix $\mathbf{C}_n \mathbf{C}_n^T$ are used replacing $\mathbf{u}_i^{\{I_m\}}$ in (2). It should be emphasized, however, that although the reconstruction procedure is the same in both methods, (2), the time-series obtained are different, since the eigenvectors employed are not the same.

C. Performance Metrics

There are some figures of merit to assess the TS, recognized in finance as performance metrics or measures.

Percentage accuracy (profitable trades to total trades) is a naive approach. More adequate measures based on profit are as follows. The Profit Factor F is defined as the ratio between gross profit and gross loss (sum of P winning trades S^+ divided by Q losing trades S^-). The Payoff Ratio $\mathcal{P}$ is computed as the ratio between the geometric mean of winning $\overline{P^+}$ and of losing trades $\overline{P^-}$. The expected value is defined as the difference between expected profits and expected costs.

The previous measures, detailed in Table I, inform the amount of money made with respect to the amount of money lost, but neglects the risk exposure. In stock markets, the prevailing approaches include some risk metrics [34] to evaluate the perceived

TABLE II
RISK-ADJUSTED PERFORMANCE METRICS

Measure	$\mathbb{S}$	Ω	$\mathcal{C}$	φ
$\mathcal{R}$	σ_R	$\Upsilon_1(\zeta)$	$-MD_1$	ν

profit, called risk-adjusted metrics, computed as

$$\frac{\overline{R} - \gamma}{\mathcal{R}} \quad (5)$$

with γ standing for the risk-free interest rate (R_f), or target profit ζ ; $\overline{R}$ is the geometrical mean of historical profit; and $\mathcal{R}$ can be several risk metrics. Without loss of generality it is assumed that $\gamma = 0$.

The risk metrics, $\mathcal{R}$, considered in this paper are shown in Table II. The Sharpe Ratio, $\mathbb{S}$, measures risk by calculating the standard deviation of returns, σ_r . However, downside and upside variability have distinct meanings to an investor. Another approach is to consider the lower partial moments, a downside risk metric $\Upsilon_1(\zeta)$, measuring risk by considering only those deviations that fall below an ex ante defined threshold, as Omega Ω . Moving on, another category of risk measures consists of using drawdowns, i.e., the reduction of one's capital after a series of losing trades, and is the difference between a relative peak in capital minus a relative trough afterwards. The maximum drawdown of an asset denotes the maximum possible loss that could have been realized in the considered period of time. The Calmar Index, $\mathcal{C}$, uses the maximum drawdown, MD_1 , as its risk measure. Finally, the value at risk, ν , describes the possible loss of an investment, which is not exceeded with a given probability of $1 - \alpha$ in a certain period [35]. The Excess VaR, φ , is such an example.

IV. PROPOSED APPROACH

The proposed approach seeks to combine several techniques in order to develop a new TS. First, both trend-following [17], [19] and oscillator [18] TTRs based on the SSA are pooled. Second, the two online implementation versions of SSA are considered, the causal [17], [18] and the recursive [20] versions. Finally, different decision-making processes are considered in order to adjust the trading strategies over time, where performance metrics are considered.

A. Overview of the Proposed Trading System

In an overview, the proposed TS is strongly related to the concept of ensembles or committees [36]. The main idea is that distinct trading strategies of individual subsystems can offer complementary information concerning a scenario of uncertainty, as occurs in stock markets. Thus, the overall performance can be improved.

In this sense, the same input is supplied to all subsystems. They receive the historical time-series of a given security as real-world data, identified as in-sample data in Fig. 1(a). Some portion of the raw data is processed according to the SSA in order to generate TIs. By adjusting some SSA parameters, such

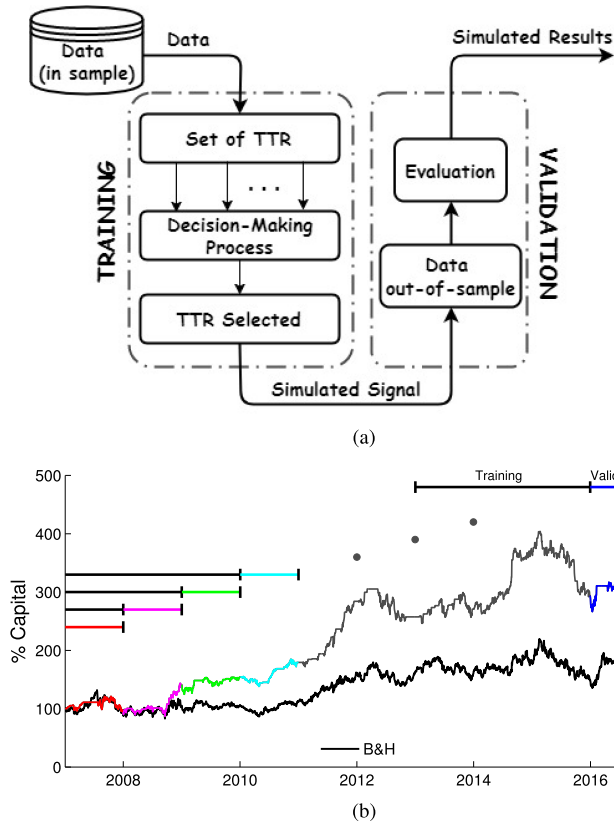


Fig. 1. Main steps for the design of the proposed TS. (a) Schematic of the design of the proposed TS. Training and validation processes occur, corresponding to one single iteration. (b) Evolution of the cumulative capital curve obtained from concatenating the results of each iteration (ranging one year) of the simulation process.

as embedding dimension and grouping, information from distinct time-scales can be analyzed. In the case of recursive SSA [20], the forgetting factor can be adjusted too. Then, each TTR operates in parallel in the remaining portion of the training data, and the individual performances are evaluated by the metrics described in Section III-C. This summarizes the step identified as “Set of TTR” in Fig. 1(a).

The next step is to combine the subsystems, which is the block labeled as “Decision-Making Process” in Fig. 1(a). There are several methods for combining the outputs of the subsystems: majority voting, weighted voting, naive Bayesian, among others. The details concerning this step will be discussed later. Therefore the block “TTR Selected” in Fig. 1(a) concludes the training stage of the proposed TS, returning a single TTR as output.

Once a TTR is selected, it is validated by out-of-sample data. By means of this approach, it is possible to mimic the result that would occur in real time, using simulated signals in the validation data of this iteration. These data are incorporated into the training data of the next iteration, a methodology commonly referred as walk-forward test, as illustrated in Fig. 1(b). In this figure, the validation and training data are indicated by colored and black bars, respectively. The B&H strategy is also presented for the same security for which the TS operates for the sake of comparison. Notice that the TS may be also evaluated using distinct performance metrics.

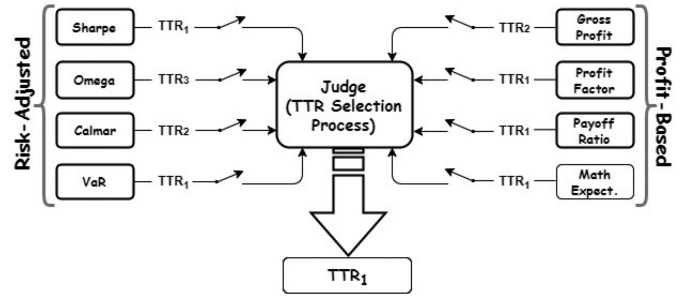


Fig. 2. A schematic diagram that illustrates the decision-making process considered in the proposed TS.

B. Decision-Making Process

An interesting feature of the proposed TS is an automated decision-making process to choose a single trading strategy from a pool of subsystems, for a given time-frame. Roughly speaking, decision-making process requires the use of some method to evaluate a list of criteria in order to take a decision. In this paper, the list of criteria is composed by different performance metrics. In the sequel, two options are discussed.

1) *Evaluation Through a Single Criterion:* This is the simplest version of the method, which requires only one of the performance metrics, shown in Tables I and II, as the decision criterion. In this case, the trader may have preference for a particular measure, committing itself to it. For the sake of this work, all TTRs are evaluated using the same performance metric, and the TTR that reaches the best (maximum) value is selected. In this context, the evaluation and the decision-making block diagram shown in Fig. 1(a) is represented by “switch ON” in a single performance metric in Fig. 2.

2) *Evaluation Through Multiple Criteria Using Ensemble:* Decision-making process that takes into account two or more criteria requires the use of MCDM. The objectives of these methods are to select, classify, sort, and learn from a group of different alternatives available [37]. When applied to stock time-series, an ensemble might allow different classifiers to potentially learn different temporary patterns among a list of candidates in the dataset. Therefore, this procedure may result in a more reliable decision.

The method for classification employed in this manuscript is the majority voting. Each performance metric “votes” on a single TTR. Each vote can be regarded as the evaluation through a single criterion, described in the previous section. Then, the final decision (represented as the “judge” block in Fig. 2) is taken based on which TTR receives more votes. For example, Fig. 2 shows the TTR “voted” by each performance metric. For instance, TTR₁ and TTR₃ were elected for the Sharpe and Omega indices, respectively. Supposing this specific ensemble decision process, the TTR₁ would be selected, because it had been voted five times, whereas TTR₂ and TTR₃ received two and one votes, respectively.

Finally, if two (or more) TTRs receive the same number of votes, an arbitrary order of priorities has been defined: Sharpe, Omega, E. Var, Calmar, Profit, P. Factor, and Payoff. For instance, if TTR₁ is elected by Omega and Payoff, while TTR₂

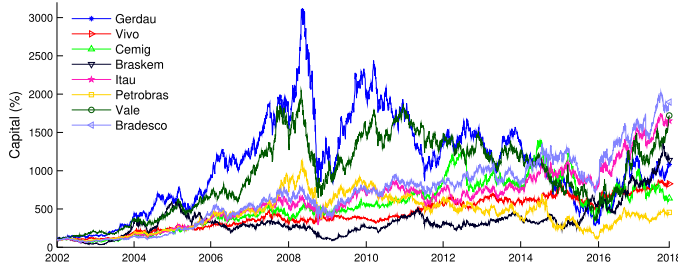


Fig. 3. Capital cumulative curves of the securities used in this paper.

is chosen by Calmar and Profit, the “judge” declares TTR_1 as the winner.

C. Remarks on the Advantages of the Proposed TS

An important feature of Ensembles is the diversity in terms of individual subsystems. Concerning the proposed TS, the diversity can be justified by some features.

For the training sets, the diversity is encouraged since different subsets of data are presented to the TS over time. Regarding input features, the same data is analyzed by means of distinct TI, also providing diversification. By resorting to the SSA, for instance, short- and long-term information can be obtained from the stock price time-series. Therefore, for the same subset of data distinct patterns are available. Finally, the output target is also diverse, since the same TTR performance may be evaluated by distinct performance metrics [37]. In other words, the same result is appraised by different eyes.

The combination of these strategies allows that the chosen set of TTRs may disagree with each other. Therefore, according to the market condition the individual failures can be minimized by changing the trading strategy, which can be very interesting from the risk-adjustment standpoint [36].

V. EXPERIMENTAL FRAMEWORK

In this section, data, parameters, and methodology used for the computational simulations are presented. The proposed TS is backtested using eight Brazilian stocks, and compared to an ensemble based on classic TA tools, henceforth called classical trading system (CTS) and the benchmark B&H strategy.

A. Data—Brazilian Stocks

The database used in this work comprises series of daily prices of some of the most prominent stocks of the main Brazilian stock market index (IBOVESPA), including: Bradesco, Itaú, Braskem, Cemig, Gerda, Vivo, Petrobras, and Vale, respectively; tickers BBDC4, ITUB4, BRKM5, CMIG4, GGBR4, VIVT4, PETR4, and VALE3. The period of analysis extends from January 3rd, 2002 to December 31st, 2017. Fig. 3 shows the capital curves. Table III presents the main statistics of the returns. These stocks are selected based on three different assumptions: 1) they are negotiated since January 2nd, 2000; 2) they are from different economic sectors;

TABLE III
STATISTICS OF THE RETURNS, $N = 3961$, ASYMMETRY (ASYM.), KUROSITS (KURT.), JARQUE-BERA (JB) TEST, β IS THE FAMA AND FRENCH [38] THREE-FACTOR MODEL (FF3) MARKET BETA AND S IS THE SHARPE RATIO

Stock	$\bar{R}(10^4)$	$\sigma(10^2)$	Asym.	Kurt.	JB	β	$S(10^2)$
BBDC4	7.431	2.211	0.473	8.093	4429	0.513*	3.361
BRKM5	6.140	2.758	0.321	6.651	2267	0.353*	2.226
CMIG4	4.691	2.539	-0.204	7.788	3811	0.408*	1.848
GGBR4	6.015	2.761	0.205	5.520	1075	0.525*	2.179
ITUB4	7.081	2.268	0.616	9.287	6775	0.541*	3.122
PETR4	3.832	2.621	0.146	6.644	2205	0.595*	1.462
VALE3	7.187	2.616	0.168	6.397	1923	0.543*	2.747
VIVT4	5.343	1.823	0.148	4.443	358.2	0.238*	2.931

* indicates p -value < 0.01 .

TABLE IV
TECHNICAL TRADING RULES BASED ON SSA INDICATORS, SSA-TTR, FOR DIFFERENT EMBEDDING DIMENSION: $L = 5; 6; \dots; 12; 13; 15; 17; \dots; 25; 27; 30; 33; 36; \dots; 57; 60; 65; 70; \dots; 95; 100; 110; 120; \dots; 200$

Type	Grouping
Trend-Following	$\{Im\} = \mathbf{A}_{\{1,2\}}, \forall L;$
	$\{Im\} = \mathbf{A}_{\{1,2,3\}}, \text{ for } L \geq 30;$
	$\{Im\} = \mathbf{A}_{\{1,2,3,4\}}, \text{ for } L \geq 100;$
	$\{Im\} = \mathbf{A}_{\{1,2,3,4,5\}} \text{ for } L \geq 160.$
Oscillators	$\{Im\} = \mathbf{A}_{\{2\}}, \forall L;$
	$\{Im\} = \mathbf{A}_{\{2,3\}}, \text{ for } L \geq 30;$
	$\{Im\} = \mathbf{A}_{\{2,3,4\}}, \text{ for } L \geq 100;$
	$\{Im\} = \mathbf{A}_{\{2,3,4,5\}} \text{ for } L \geq 160.$

and 3) their particular weights to the IBOVESPA index are relevant.¹

B. Technical Indicators and Trading Rules Investigated

Table IV shows the SSA-TTRs, and Table V shows the classic TTRs, selected for this work as subsystems for the proposed TS and the CTS, respectively. Given the set of parameters in these tables, there is a total of 270 rules for each approach. A description regarding SSA trending-following rules and moving averages are presented in [17]. Most of the classic oscillators and the SSA-TTRs are detailed in [18]. The Bollinger Bands [39] are defined as follows. Three series are calculated from the price: a central line (generally, a moving average); and two bands in which a multiplicative factor based in the standard deviations are added (superior band) or subtracted (inferior band), according to

$$\begin{cases} CL_n[L] = MA_n[L] \\ HB_n[\eta, L] = CL_n[L] + \eta\sigma_n \\ LB_n[\eta, L] = CL_n[L] - \eta\sigma_n \end{cases} \quad (6)$$

¹This can be confirmed by the value of β . This parameter was calculated using the Fama–French three-factor model [38], in which the risk factors for Brazil were obtained at <http://www.nefin.com.br>. The dataset are formed by daily closing prices and the period of analysis extends from January 3rd, 2002 to December 31st, 2017.

TABLE V
TECHNICAL TRADING RULES BASED ON CLASSIC INDICATORS, CLASSIC-TTR

Indicators	Parameters
Moving Averages	$L = \begin{cases} 5, 6, \dots, 12, \\ 13, 15, 17, \dots, 25, \\ 27, 30, 33, 36, \dots, 57, \\ 60, 65, 70, \dots, 95, \\ 100, 110, 120, \dots, 200. \end{cases}$
Bollinger Bands	$L = 18, 19, \dots, 22;$
	$\eta = 1.8, 1.85, \dots, 2.2.$
MACD	$P = 11, 12, 13;$
	$Q = 24, 25, \dots, 28;$
	$K = 8, 9, 10.$
Momentum	$K = 3, 4, 5, \dots, 52.$
RSI	$L = 12, 13, \dots, 16;$
	$RSI^- = 25, 30, 35;$
	$RSI^+ = 65, 70, 75.$
Stochastic	$M = 8, 11, 14, 17;$
	$L = 5, 8, 11, 14;$
	$D^- = 25, 30; D^+ = 80, 85.$

where $CL_n[L]$ indicates the central line; $HB_n[\eta, L]$ and $LB_n[\eta, L]$ represent the superior and inferior band, respectively, calculated at instant n using a period of L days; σ_n stands for the standard deviation; and η is the multiplicative factor to be used. Different approaches have been applied using Bollinger Bands [39] for trading rules design. In this work, a buy signal is triggered when the price crosses the inferior band downwardly, whereas a close signal is issued at the moment in which the price crosses the inferior band upwardly.

Finally, as a simulation detail, recursive SSA was implemented solely for the trend-following indicators. This was necessary to ensure that TTRs based on SSA and on classic tools are equally present, as well as to equalize between trend-following and oscillator TTR types. It was used as (*ad hoc*) decay $\varphi = 0.95$. Simulations were carried out using slight changes ($0.92 \leq \varphi \leq 0.98$) but similar results were found, and are omitted for the sake of brevity.

C. Test Methodology

Each of the SSA-based TTR described in Table IV was combined to all performance metrics in Table I. Therefore, since there are 270 TTRs and eight metrics, there is a total of 2160 evaluations of the proposed TS. To establish a fair comparison, the same number of classic TTR is considered for the pool of the CTS. The whole dataset was divided in three subperiods: 2007–2010, 2011–2014, and 2015–2017, representing the beginning of new presidential mandates in Brazil. Therefore, four time-frames are considered in the simulations: the complete period (2007–2017) and the three subperiods. Only long positions were

TABLE VI
DIFFERENCE BETWEEN THE AVERAGE RETURN FOR BOTH TS ($\bar{R}_{TS}$) AND THE B&H STRATEGY ($\bar{R}_{BH}$), GIVEN BY $\Delta = (\bar{R}_{TS} - \bar{R}_{BH}) 10^3$

Tick	Method	2007–2010		2011–2014		2015–2017		2007–2017	
		Δ	t-stat	Δ	t-stat	Δ	t-stat	Δ	t-stat
BBDC4	SSA	-0.417	-0.596	-0.178	-0.453	-0.255	-0.433	-0.290	-0.872
	CTS	-1.095	-1.758	-0.611	-1.331	-0.691	-1.233	-0.808	-2.526
BRKM5	SSA	-0.048	-0.064	0.375	0.685	-0.549	-0.726	-0.028	-0.072
	CTS	-0.433	-0.549	-0.280	-0.428	-0.859	-0.967	-0.489	-1.102
CMIG4	SSA	-0.078	-0.172	-0.453	-0.994	1.222	1.393	0.149	0.446
	CTS	-0.874	-1.600	-0.094	-0.187	0.722	0.659	-0.156	-0.386
GGBR4	SSA	0.253	0.326	0.510	0.971	-0.313	-0.356	0.210	0.505
	CTS	-0.382	-0.529	0.579	0.998	-1.220	-1.281	-0.242	-0.568
ITUB4	SSA	-0.068	-0.122	-0.571	-1.359	-0.160	-0.316	-0.268	-0.927
	CTS	-1.016	-1.590	-0.668	-1.365	-0.923	-1.826	-0.856	-2.649
PETR4	SSA	-0.090	-0.130	0.327	0.534	0.357	0.382	0.208	0.493
	CTS	-0.322	-0.484	0.193	0.303	-0.649	-0.704	-0.199	-0.475
VALE3	SSA	-0.551	-0.744	0.160	0.327	-0.619	-0.638	-0.300	-0.720
	CTS	-0.782	-1.035	-0.213	-0.427	-0.390	-0.459	-0.478	-1.190
VIVT4	SSA	-0.323	-0.769	0.346	0.932	0.019	0.050	0.023	0.098
	CTS	-0.772	-1.730	-0.741	-2.280	-0.336	-0.834	-0.642	-2.807

Following [8], all t -statistics are compared with 1.95, the critical t -value for two-tailed test at 5% level for large numbers of observations.

considered in the results presented. Also, when the TS emits a sell signal the trader does not invest the capital elsewhere.

These experiments investigate whether the proposed ex ante, ensemble TS based on the SSA, is able to outperform a pool of classic tools from the realm of TA, the CTS, and also the B&H, concerning profitability and risk-adjusted performances. These experiments do not intend to commit neither with the EMH nor AMH, but to provide a more systematic, yet simplistic, framework that can be used by investors and practitioners.

The question of predictability power of the TS and the CTS is tested following the methodology described in [8, Section 3.2]. This test verifies whether the mean returns of the proposed TS are different than the mean return of the B&H method. The test is also carried out for the CTS. Since returns cannot be properly evaluated in the absence of its attendant risk [34], a risk-adjust analysis is also performed. In this sense, an ex post analysis is conducted considering the performance measures described in Section III-C. Whether a single metric is considered, the performance is evaluated according to that metric alone. Alternatively, whether the ensemble is considered, the performance is evaluated for all the metrics.

VI. RESULTS

In the sequel, results of the comparison among the proposed TS, the CTS, and the benchmark B&H strategy are presented.

A. Results by Multiple Criteria Using Ensemble

In this section, the decision-making process described in Section IV-B.2 is considered. Table VI presents the difference between the average return for the proposed TS ($\bar{R}_{TS}$) and the B&H strategy ($\bar{R}_{BH}$), according to

$$\Delta = (\bar{R}_{TS} - \bar{R}_{BH}). \quad (7)$$

TABLE VII
RISK METRICS FOLLOWING NOMENCLATURE ADOPTED IN TABLE II

Tick	Year	σ_R (%)			$\Upsilon_1(\zeta)$ (10^3)			$-MD_1$ (%)			ν (10^2)		
		SSA	BH	CTS	SSA	BH	CTS	SSA	BH	CTS	SSA	BH	CTS
BBDC4	2007–2010	1.40	2.61	1.72	3.56	8.85	5.12	31.07	52.44	61.66	2.34	4.36	2.79
	2011–2014	1.38	1.85	1.16	3.37	6.66	2.78	28.37	26.09	31.15	2.30	3.10	1.89
	2015–2017	1.42	2.14	1.50	3.60	7.41	3.78	27.51	44.33	35.51	2.40	3.61	2.49
	2007–2017	1.40	2.23	1.48	3.51	7.66	3.91	31.07	52.44	67.77	2.34	3.73	2.41
BRKM5	2007–2010	1.72	2.91	1.53	4.15	10.02	3.47	44.72	75.76	42.44	2.91	4.87	2.56
	2011–2014	1.54	2.31	1.05	4.04	8.43	2.38	27.15	56.21	41.77	2.60	3.83	1.73
	2015–2017	1.94	2.82	1.46	4.27	9.00	2.42	38.82	40.55	23.22	3.32	4.83	2.51
	2007–2017	1.72	2.68	1.36	4.14	9.16	2.78	44.72	75.76	42.44	2.92	4.50	2.27
CMIG4	2007–2010	1.66	2.18	1.34	4.15	7.77	3.03	41.87	34.67	55.22	2.77	3.64	2.18
	2011–2014	1.51	2.08	1.34	3.46	7.02	3.27	42.51	43.22	30.46	2.51	3.49	2.27
	2015–2017	2.20	3.25	1.29	5.19	12.03	1.63	35.46	72.18	21.97	3.73	5.34	2.20
	2007–2017	1.77	2.48	1.33	4.18	8.66	2.74	42.51	75.37	55.22	2.97	4.13	2.22
GGBR4	2007–2010	1.96	3.13	2.16	4.90	11.18	5.59	34.73	74.50	55.55	3.33	5.23	3.60
	2011–2014	1.40	2.16	1.16	3.68	8.56	2.72	27.06	64.73	24.95	2.29	3.50	1.91
	2015–2017	2.45	3.42	2.23	6.31	12.47	5.60	36.16	70.03	45.45	4.10	5.73	3.65
	2007–2017	1.93	2.91	1.88	4.84	10.59	4.55	56.64	90.90	64.50	3.24	4.82	3.11
ITUB4	2007–2010	2.27	2.87	2.05	5.10	9.37	6.08	36.86	56.36	65.61	3.81	4.81	3.36
	2011–2014	1.33	1.88	1.07	3.75	6.95	2.88	39.30	36.27	45.86	2.17	3.12	1.73
	2015–2017	1.44	1.99	1.44	3.29	6.73	3.68	23.61	33.55	31.83	2.45	3.37	2.37
	2007–2017	1.75	2.31	1.58	4.11	7.77	4.26	42.97	56.36	72.11	2.93	3.87	2.59
PETR4	2007–2010	1.68	2.75	1.78	4.05	9.44	4.42	41.44	67.83	45.18	2.81	4.58	2.96
	2011–2014	1.56	2.48	1.47	4.08	9.27	3.80	40.51	62.87	44.62	2.54	4.03	2.38
	2015–2017	2.30	3.43	2.35	4.81	11.93	6.13	29.92	70.79	44.60	3.96	5.78	3.93
	2007–2017	1.83	2.86	1.85	4.27	10.08	4.66	50.12	89.57	54.05	3.07	4.75	3.06
VALE3	2007–2010	1.92	3.02	1.86	4.99	10.21	4.95	42.52	68.52	60.38	3.21	5.07	3.09
	2011–2014	1.14	1.91	1.10	2.88	7.60	3.05	49.90	61.34	59.00	1.84	3.09	1.73
	2015–2017	2.34	3.52	2.66	5.81	12.58	6.57	49.26	67.58	53.05	3.95	5.96	4.50
	2007–2017	1.82	2.83	1.91	4.44	9.91	4.71	62.93	82.93	83.32	3.03	4.72	3.15
VIVT4	2007–2010	1.42	1.94	1.34	4.02	7.03	3.77	49.39	37.03	56.79	2.34	3.22	2.16
	2011–2014	1.06	1.58	1.21	2.27	5.69	3.59	14.73	22.62	38.04	1.84	2.66	1.97
	2015–2017	1.21	1.60	1.17	3.16	5.79	3.20	15.49	38.41	32.22	2.04	2.68	1.93
	2007–2017	1.24	1.72	1.25	3.15	6.21	3.55	49.39	38.41	68.64	2.09	2.88	2.03

The same comparison is presented for the CTS. The t -statistics is also reported. The results are shown for the entire period and for the three subperiods.

By analyzing Table VI, one can observe that the differences of return between the B&H strategy and the TS (both for the proposed approach and for classic TA tools) are not statistically significant. A few number of exceptions can be viewed for the CTS in securities BBDC4, ITUB4, and VIVT4. Nevertheless, TS returns are inferior to B&H strategy in these cases.

Table VII reports a risk analysis for the abovementioned simulations. Four different measures are used. It is worth mentioning that the risk associated with the proposed TS is consistently lower than those of B&H. Exceptions occur only in five cases, all for the maximum drawdown: in subperiod 2011–2014, for stocks BBDC4 and ITUB4; in subperiod 2007–2010, for stocks VIVT4 and CMIG4; and in all period for stock VIVT4. Comparing the proposed TS and CTS, there are mixed results. In 128 simulations (i.e., for all securities, metrics, and periods), the proposed TS provides lower risks in 63 situations; whereas the CTS, in 65.

The risk-adjusted performance metrics, computed according to (5), are presented in Table VIII. As can be seen by analyzing this table, considering the entire period of simulation (2007–2017), one can observe that the proposed TS presents a superior performance to classic TA tools in almost all simulations (57 of 64): in all metrics for GGBR4 and PETR4; in seven (all but P. Factor) metrics for BBDC4, BRKM5, ITUB4, VIVT4, and VALE3; and six for CMIG4. Moreover, the results

of SSA-TTR are positive in all simulations, whereas classic-TTR were negative in 36 simulations: for BBDC4, GGBR4, ITUB4, PETR4, VALE3, and VIVT4 in all metrics except the Profit Factor and Payoff.

In Table VIII, the performance metrics for the Buy & Hold strategy is also presented when possible.² Considering the entire period of simulation (2007–2017), the proposed TS outperforms B&H in 32 of 40 simulations: in all metrics for BRKM5, CMIG4, GGBR4, PETR4, and VIVT4; in seven metrics (all except Profit) for BBDC4; in six metrics for VALE3; and in four metrics for ITUB4. Moreover, the proposed TS reached to positive results in all simulations, whereas B&H results are negative for GGBR4 and PETR4 in all metrics, totaling 10 simulations. Regarding the subperiods analysis, similar observations can be made, providing qualitatively resembling results.

B. Results by a Single Performance Metric

Risk-adjusted performance metrics, computed according to (5), are presented in Table IX, as described in the previous section; however, the decision-making process follows Section IV-B.1 this time, instead of the one from Section IV-B.2.³ Based on the analysis of Table IX, considering the entire period of simulation (2007–2017), one can notice that SSA-TTR outperforms classic-TTR in 42 out of 64 simulations: in seven metrics for VIVT4; in six metrics for VALE3 and BBDC4; in five metrics for ITUB4, GGBR4, and PETR4; and in four metrics for BRKM4 and CMIG4. Also, it can be noticed that SSA-TTR results are negative along only in simulations: for BBDC4, GGBR4, and PETR4 in expectation metric. In contrast, the classic-TTR results are negative⁴ along 26 simulations: in six metrics for VIVT4; in five metrics for ITUB4; in four metrics BBDC4, GGBR4, and VALE3; and in three metrics for PETR4.

C. Ensembles Comparison: Risk-Adjusted Versus Profit-Based

As discussed in Section IV-B.2, several combinations of performance metrics can be merged providing different sorts of ensembles. Two particular ensembles are compared: one based on risk-adjusted performance metrics, whereas the other is constructed using profit-based ones.

Analysis of Table X reveals that the risk-adjusted ensemble outperforms the profit-based counterpart in 70% of the simulations. Moreover, all simulations regarding risk-adjusted ensemble present positive results, whereas profit-based ensemble results are negative in some situations. The risk-adjusted ensemble also surpass the B&H in 77% experiments, whose negative results occurred for GGBR4 and PETR4. Finally, the profit-based ensemble surpassed the B&H in 20% and underperformed in 70% of the simulations. Inconclusive results have been reported for GGBR4 and PETR4, in Omega ratio; for BRKM5, in Calmar ratio; and for GGBR4, in Profit (both negatives).

²The Buy & Hold strategy executes only one trade per period, thereby some metrics (Profit, Payoff, and Expectation) cannot be calculated.

³An analysis similar to the ones carried out in Tables VI and VII is omitted because of the lack of space.

⁴For Omega ratio, a value lower than one is considered negative.

TABLE VIII
RESULTS FOR A DECISION-MAKING PROCESS USING AN ENSEMBLE ENCOMPASSING ALL PERFORMANCE METRICS

Tick	Year	Sharpe (10^2)			Omega			E. VaR (10^2)			Calmar (10^2)			Profit (%)			Payoff		P. Factor		Expect.	
		SSA	B&H	CTS	SSA	B&H	CTS	SSA	B&H	CTS	SSA	B&H	CTS	SSA	B&H	CTS	SSA	CTS	SSA	CTS	SSA	CTS
BBDC4	2007–2010	1.654	1.584	-2.880	1.065	1.047	0.903	0.991	0.946	-1.773	0.075	0.079	-0.080	25.71	50.20	-38.63	1.735	0.710	2.015	1.516	0.371	-0.221
	2011–2014	1.382	1.579	-1.846	1.057	1.044	0.923	0.830	0.945	-1.131	0.067	0.112	-0.069	20.82	33.61	-19.11	1.358	0.658	0.412	1.287	0.077	-0.238
	2015–2017	3.996	3.244	0.786	1.157	1.094	1.031	2.362	1.922	0.474	0.206	0.156	0.033	52.05	67.01	9.11	2.641	1.157	0.401	2.032	0.208	0.070
	2007–2017	2.143	1.982	-1.554	1.086	1.058	0.941	1.281	1.183	-0.950	0.097	0.084	-0.034	125.95	231.82	-46.38	1.725	0.868	0.504	1.662	0.151	-0.113
BRKM5	2007–2010	3.280	1.181	1.377	1.136	1.034	1.061	1.945	0.706	0.827	0.127	0.045	0.050	74.65	40.35	23.15	2.608	1.424	3.238	1.620	0.766	0.176
	2011–2014	3.215	-0.122	-0.920	1.123	0.997	0.959	1.909	-0.074	-0.561	0.183	-0.005	-0.023	63.36	-2.76	-9.15	6.766	0.776	2.674	1.869	1.624	-0.180
	2015–2017	5.845	5.205	6.179	1.265	1.163	1.374	3.412	3.042	3.605	0.292	0.362	0.389	131.18	196.38	95.24	4.201	10.210	1.524	10.089	0.747	4.544
	2007–2017	4.018	1.902	2.119	1.167	1.056	1.103	2.372	1.134	1.267	0.155	0.067	0.068	555.67	300.05	118.43	3.615	1.975	2.230	2.308	0.876	0.434
CMIG4	2007–2010	2.008	1.421	-3.111	1.080	1.040	0.862	1.200	0.851	-1.920	0.079	0.089	-0.076	38.78	35.73	-33.80	1.489	0.582	0.414	1.873	0.100	-0.347
	2011–2014	0.994	2.399	3.983	1.043	1.071	1.163	0.598	1.428	2.355	0.035	0.115	0.175	16.00	63.86	69.68	1.285	1.972	0.419	1.958	0.048	0.455
	2015–2017	4.357	-1.719	4.744	1.184	0.954	1.376	2.564	-1.045	2.793	0.270	-0.077	0.280	103.06	-33.85	57.51	2.211	15.049	1.114	7.462	0.359	4.641
	2007–2017	2.516	0.573	1.579	1.107	1.016	1.077	1.499	0.344	0.947	0.105	0.019	0.038	235.82	47.24	76.93	1.632	1.537	0.535	1.891	0.132	0.248
GGBR4	2007–2010	4.426	1.011	0.875	1.177	1.028	1.034	2.605	0.605	0.526	0.250	0.043	0.034	135.26	36.64	20.52	2.983	1.228	2.827	1.565	0.914	0.090
	2011–2014	-1.228	-3.778	-0.620	0.953	0.905	0.974	-0.749	-2.335	-0.377	-0.064	-0.126	-0.029	-15.65	-55.51	-6.88	0.468	0.930	1.138	1.373	-0.389	-0.068
	2015–2017	1.793	1.371	-1.882	1.069	1.038	0.925	1.071	0.818	-1.149	0.121	0.067	-0.092	38.32	41.43	-26.67	2.360	0.583	2.877	1.519	0.662	-0.322
	2007–2017	1.960	-0.229	-0.342	1.078	0.994	0.986	1.171	-0.138	-0.207	0.067	-0.007	-0.010	180.05	-16.59	-16.03	2.229	1.037	2.478	1.527	0.581	-0.012
ITUB4	2007–2010	2.422	1.618	-1.737	1.108	1.050	0.941	1.442	0.966	-1.061	0.149	0.082	-0.054	71.86	58.03	-29.63	1.671	0.913	0.525	1.290	0.150	-0.141
	2011–2014	-2.231	0.995	-3.382	0.921	1.027	0.874	-1.369	0.598	-2.092	-0.076	0.052	-0.079	-25.49	20.31	-30.18	0.758	0.569	0.346	1.345	-0.088	-0.316
	2015–2017	4.857	3.844	-0.440	1.213	1.114	0.983	2.856	2.270	-0.267	0.296	0.228	-0.020	67.77	76.16	-4.59	2.456	1.006	0.433	2.369	0.210	-0.009
	2007–2017	1.610	1.893	-1.759	1.069	1.056	0.935	0.964	1.130	-1.076	0.066	0.078	-0.039	115.47	228.59	-53.12	1.457	0.825	0.425	1.614	0.093	-0.174
PETR4	2007–2010	2.160	0.794	0.635	1.089	1.023	1.026	1.289	0.476	0.382	0.088	0.032	0.025	43.00	23.99	11.80	1.737	1.188	1.412	1.698	0.254	0.079
	2011–2014	-2.075	-3.376	-3.036	0.921	0.910	0.883	-1.272	-2.080	-1.872	-0.080	-0.133	-0.100	-27.41	-56.35	-35.63	0.285	0.419	0.805	1.627	-0.549	-0.475
	2015–2017	6.133	2.135	1.676	1.294	1.061	1.064	3.570	1.268	1.002	0.472	0.104	0.088	184.18	72.01	33.80	7.182	1.663	3.885	2.312	2.053	0.333
	2007–2017	2.151	-0.193	-0.096	1.092	0.995	0.996	1.284	-0.117	-0.058	0.079	-0.006	-0.003	191.85	-13.97	-4.73	2.145	1.115	1.966	1.872	0.447	0.025
VALE3	2007–2010	1.790	2.074	0.669	1.069	1.061	1.025	1.071	1.234	0.403	0.081	0.091	0.021	40.39	85.31	13.08	1.647	1.256	1.560	1.721	0.280	0.101
	2011–2014	-3.925	-3.793	-7.425	0.844	0.904	0.733	-2.436	-2.346	-4.711	-0.090	-0.118	-0.138	-35.85	-51.28	-55.45	0.313	0.393	0.240	1.114	-0.318	-0.462
	2015–2017	3.111	2.836	3.288	1.125	1.079	1.133	1.843	1.677	1.945	0.148	0.148	0.165	71.28	109.47	91.10	2.269	1.785	0.310	2.301	0.141	0.359
	2007–2017	0.936	0.828	-0.130	1.038	1.024	0.995	0.563	0.496	-0.078	0.027	0.028	-0.003	58.83	89.12	-6.50	1.432	1.133	0.645	1.939	0.104	0.025
VIVT4	2007–2010	-0.762	0.660	-4.082	0.973	1.018	0.855	-0.463	0.397	-2.534	-0.022	0.035	-0.096	-10.14	13.45	-41.66	0.921	0.391	0.324	1.100	-0.031	-0.458
	2011–2014	8.472	3.067	-1.687	1.396	1.085	0.943	4.884	1.822	-1.032	0.612	0.215	-0.054	143.86	61.68	-18.27	3.032	0.623	0.786	1.500	0.414	-0.286
	2015–2017	3.232	1.979	0.347	1.124	1.055	1.013	1.920	1.183	0.210	0.253	0.082	0.013	33.58	26.41	3.04	2.109	1.150	0.687	1.447	0.265	0.064
	2007–2017	3.169	1.739	-2.173	1.125	1.048	0.924	1.883	1.041	-1.334	0.080	0.078	-0.039	191.76	126.01	-52.11	1.846	0.624	0.540	1.314	0.182	-0.271

A comparative study of the proposed approach is made against a TS based on classic indicators (CTS) as well as the benchmark B&H strategy.

TABLE IX
RESULTS FOR A DECISION-MAKING PROCESS IN WHICH A SINGLE CRITERION IS USED BASED ON EACH PERFORMANCE METRIC TAKEN SEPARATELY

Tick	Year	Sharpe (10^2)			Omega			E. VaR (10^2)			Calmar (10^2)			Profit (%)			Payoff		P. Factor		Expect.	
		SSA	B&H	CTS	SSA	B&H	CTS	SSA	B&H	CTS	SSA	B&H	CTS	SSA	B&H	CTS	SSA	CTS	SSA	CTS	SSA	CTS
BBDC4	2007–2010	1.654	1.584	-2.420	1.008	1.047	0.963	0.991	0.946	-1.485	0.052	0.079	0.018	34.13	50.20	-36.25	0.830	2.285	1.188	2.118	0.006	-0.098
	2011–2014	1.382	1.579	-1.846	1.057	1.044	0.909	0.830	0.945	-1.131	0.103	0.112	-0.029	40.31	33.61	-15.16	1.705	2.166	3.151	0.869	-0.305	0.066
	2015–2017	3.996	3.244	0.786	1.157	1.094	1.031	2.362	1.922	0.474	0.203	0.156	0.033	52.05	67.01	9.11	1.151	—	2.454	4.209	0.216	0.635
	2007–2017	2.143	1.982	-1.376	1.054	1.058	0.965	1.281	1.183	-0.840	0.099	0.084	0.007	179.95	231.82	-41.57	0.942	2.332	2.045	1.749	-0.035	0.018
BRKM5	2007–2010	3.280	1.181	1.377	1.191	1.034	1.448	1.945	0.706	0.827	0.385	0.045	0.194	74.65	40.35	40.50	0.255	3.670	0.648	5.290	-0.057	0.420
	2011–2014	3.215	-0.122	-0.920	1.090	0.997	0.871	1.909	-0.074	-0.561	0.128	-0.005	-0.001	63.36	-2.76	-17.37	0.190	1.991	1.184	0.690	-0.006	-0.113
	2015–2017	6.312	5.205	6.179	1.012	1.163	1.335	3.674	3.042	3.605	0.363	0.362	0.547	154.54	196.38	93.91	0.979	4.006	1.257	6.696	0.300	0.760
	2007–2017	4.182	1.902	2.119	1.101	1.056	1.215	2.467	1.134	1.267	0.220	0.067	0.147	621.91	300.05	125.14	0.308	2.898	0.978	3.189	0.009	0.303
CMIG4	2007–2010	2.008	1.421	-3.111	1.080	1.040	1.000	1.200	0.851	-1.920	0.141	0.089	-0.024	51.05	35.73	14.69	1.477	0.000	3.681	0.000	0.060	-0.593
	2011–2014	0.994	2.399	3.425	1.043	1.071	1.230	0.598	1.428	2.032	-0.017	0.115	0.123	16.00	63.86	55.18	—	7.050	1.463	14.272	—	4.367
	2015–2017	4.357	-1.719	4.744	1.184	0.954	1.280	2.564	-1.045	2.793	0.159	-0.077	0.151	-31.54	-33.85	32.57	0.619	1.859	0.742	0.853	-0.191	0.429
	2007–2017	2.516	0.573	1.359	1.107	1.016	1.253	1.499	0.344	0.816	0.066	0.019	0.055	23.24	47.24	136.13	1.212	2.166	1.900	1.261	0.039	0.159
GGBR4	2007–2010	4.426	1.011	0.875	1.177	1.028	1.021	2.605	0.605	0.526	0.365	0.043	-0.052	135.26	36.64	77.57	0.487	0.000	0.609	0.838	-0.073	1.448
	2011–2014	-1.228	-3.778	-0.620	0.944	0.905	0.920	-0.749	-2.335	-2.431	-0.020	-0.126	-0.029	-3.95	-55.51	-16.40	0.119	1.046	0.558	0.421	-0.073	-0.527
	2015–2017	1.793	1.371	-1.816	1.069	1.038	0.868	1.071	0.818	-1.149	0.186	0.067	-0.092	58.03	41.43	-26.67	1.435	3.303	3.588	5.953	-0.590	1.582
	2007–2017	1.960	-0.229	-1.022	1.075	0.994	0.956	1.171	-0.138	-0.622	0.110	-0.007	-0.036	264.35	-16.59	11.06	0.469	2.763	0.889	1.345	-0.034	0.943
ITUB4	2007–2010	1.221	1.618	-1.737	1.071	1.050	0.984	0.732	0.966	-1.061	0.094	0.082	-0.028	26.74	58.03	-9.17	0.693	1.671	2.066	0.000	0.026	0.164
	2011–2014	-2.231	0.995	-3.005	0.961	1.027	0.896	-1.369	0.598	-1.855	-0.080	0.052	-0.094	-22.36	20.31	-44.70	1.303	1.107	1.142	0.835	-0.215	-0.097
	2015–2017	4.857	3.844	-0.440	1.213	1.114	1.025	2.856	2.270	-0.267	0.321	0.228	-0.020	67.77	76.16	-4.59	0.759	—	2.090	0.000	0.407	—
	2007–2017	1.048	1.893	-1.653	1.066	1.056	0.972	0.630	1.130	-1.010	0.044	0.078	-0.020	65.56	228.59	-52.07	0.791	1.423	1.684	3.906	0.035	0.302
PETR4	2007–2010	2.160	0.794	0.321	1.089	1.023	1.008	1.289	0.476	0.194	0.087	0.032	0.097	4.0	23.99	-4.32	0.259	1.924	1.282	4.343	0.004	0.176
	2011–2014	-2.075	-3.376	-3.036	0.921	0.910	0.829	-1.272	-2.080	-1.872	-0.072	-0.133	-0.071	0.13	-56.35	-31.73	0.156	1.698	0.071	0.803	-0.368	0.022
	2015–2017	6.133	2.135	1.076	1.294	1.061	0.883	3.570	2.688	1.002	0.395	0.104	0.088	184.18	72.01	58.30	6.099	1.844	3.992	8.431	0.475	0.923
	2007–2017	2.151	-0.193	-0.202	1.092	0.995	0.986	1.284	-0.117	-0.123	0.082	-0.006	0.025	182.65	-13.97	2.31	0.274	1.887	1.029	3.051	-0.059	0.257
VALE3	2007–2010	1.790	2.074	-1.048	1.069	1.061	1.012	1.071	1.234	-0.638	0.060	0.091	0.143	16.04	85.31	98.46	0.762	2.580	2.428	2.545	0.031	0.790
	2011–2014	-3.925	-3.793	-7.425	0.856	0.904	0.842	-2.436	-2.346	-4.711	-0.106	-0.118	-0.145	-48.39	-51.28	-67.22	1.308	1.535	0.509	0.000	-0.176	-0.058
	2015–2017	3.111	2.836	3.288	1.114	1.079	1.132	1.843	1.677	1.945	0.104	0.148	0.029	71.28	109.47	-12.79	0.794	5.191	2.768	1.833	0.807	0.441
	2007–2017	0.936	0.828	-0.688	1.043	1.024	1.006	0.563	0.496	-0.418	0.005	0.028	-0.017	5.61	89.12	-43.27	0.968	2.423	1.612	1.552	0.045	0.374
VIVT4	2007–2010	-0.762	0.660	-0.482	1.156	1.018	0.806	-0.463	0.397	-2.534	-0.020	0.035	-0.070	118.78	13.45	-41.98	1.040	0.417	0.854	1.056	-0.072	-0.592
	2011–2014	8.472	3.067	-1.687	1.279	1.085	0.945	4.884	1.822	-1.032	0.367	0.215	-0.054	168.87	61.68	-27.15	2.751	3.823	5.178	2.882	1.384	0.332
	2015–2017	2.322	1.979	0.347	0.992	1.055	0.946	1.920	1.183	0.210	0.029	0.082	-0.053	37.55	26.41	16.95	1.982	2.824	2.184	5.536	0.621	0.722
	2007–2017	3.169	1.739	-2.173	1.150	1.048	0.905	1.883	1.041	-1.334	0.046	0.078	-0.041	706.43	126.01	-51.81	1.912	1.933	1.855	1.374	0.202	-0.036

TABLE X
COMPARISON OF TWO DIFFERENT ENSEMBLES: THE FIRST ONE USING RISK-ADJUSTED METRICS (LEFT SIDE OF FIG. 2),
THE SECOND ONE USING PROFIT-BASED METRICS (RIGHT SIDE OF FIG. 2)

Tick	Sharpe (10^2)			Omega			E. VaR (10^2)			Calmar (10^3)			Profit (%)			Payoff		P. Factor		Expect.	
	R-A	B&H	P-B	R-A	B&H	P-B	R-A	B&H	P-B	R-A	B&H	P-B	R-A	B&H	P-B	R-A	P-B	R-A	P-B	R-A	P-B
BBDC4	2.143	1.982	0.496	1.086	1.058	1.019	1.281	1.183	0.299	0.966	0.842	0.187	125.95	231.82	27.10	0.504	0.733	1.725	1.166	0.151	0.047
BRKM5	4.182	1.902	0.040	1.174	1.056	1.002	2.467	1.134	0.024	1.637	0.673	0.010	631.03	300.05	2.03	2.736	0.271	3.912	1.406	1.076	0.023
CMIG4	2.516	0.573	1.368	1.107	1.016	1.057	1.499	0.344	0.821	1.049	0.189	0.417	235.82	47.24	83.29	0.535	0.900	1.632	1.748	0.132	0.223
GGBR4	1.960	-0.229	-1.084	1.078	0.994	0.957	1.171	-0.138	-0.660	0.669	-0.073	-0.265	180.05	-16.59	-43.19	2.478	0.382	2.229	1.025	0.581	-0.031
ITUB4	1.048	1.893	1.140	1.044	1.056	1.047	0.630	1.130	0.684	0.426	0.777	0.483	64.47	228.59	72.07	0.376	0.569	1.327	1.438	0.063	0.110
PETR4	2.151	-0.193	-0.594	1.092	0.995	0.975	1.284	-0.117	-0.360	0.786	-0.062	-0.151	191.85	-13.97	-27.20	1.966	0.318	2.145	1.174	0.447	-0.008
VALE3	0.936	0.828	-0.189	1.038	1.024	0.993	0.563	0.496	-0.114	0.271	0.283	-0.053	58.83	89.12	-9.63	0.645	0.748	1.432	1.134	0.104	0.007
VIVT4	2.976	1.739	3.439	1.116	1.048	1.135	1.770	1.041	2.040	0.751	0.781	1.480	173.95	126.01	225.58	0.548	1.072	1.810	2.202	0.179	0.359

R-A stands for risk-adjusted ensemble, while P-B for profit-based ensemble.

D. Effects of Transaction Costs

Every negotiation in the stock market is affected by transaction costs (TC), which may have some harmful effects on the trading profits. TC encompasses: cost of commissions, trading fee, slippage, and liquidity [40]. These costs are hard to predict because these depend on the market type and on the broker. Simulations regarding this issue are discussed next using two approaches. In the first, the TC are applied in the TTR selection procedure, while in the other a break-even analysis is carried out.

1) *TC Applied in TTR Selection Procedure*: A slightly different procedure for the ensemble detailed in Section VI-A, in which all performance metrics were used in the decision process, is adopted. Using in-sample data, SSA-TTR takes into account different values for the TC (0.1%, 0.2%, and 0.3%). Then, in the validation process, a constant TC of 0.2% is applied to the final result, which means an expected cost of approximately 0.1% per trade.⁵

Four simulations were carried out using different values for trading costs during the training period, in which $TC = \{0\%, 0.1\%, 0.2\%, 0.3\%\}$. For each of the TTR selected in these four simulations, two experiments were conducted in the validation data. In the first one, no TC were applied to the validation data. Conversely, on the second one, a constant TC of 0.2% were taken into account. Table XI depicts the Sharpe ratio results for all these simulations. This table presents the B&H metric along with the TC effects whether the TC were disregarded, as detailed in the four columns referred to as “TC Neglected,” or when a constant TC of 0.2% were considered, as reported in the four columns referred to as “TC=0.2%.” Similar results are found for the other performance metrics, but are omitted for the sake of brevity.

Analysis of Table XI indicates that the inclusion of TC during the decision-making process (in-sample data) affects the trading outcome. However, this does not happen in all simulations.

If no TC were applied to the validation data, the SSA-TTR surpasses the B&H in 25 out of 32 simulations. If a constant TC of 0.2% were applied, SSA-TTR still outperforms the B&H in 20 simulations. In two conditions, for BBDC4 there is a draw. It must be stressed, nonetheless, that SSA-TTRs present only positive results in all experiments, whereas for the B&H two negative out of eight results were reported.

⁵This value is used as a proxy because, in Brazil the stock exchange charges a fee of 0.0345% and brokers a fee of (approximately) 0.05%.

TABLE XI
EFFECTS OF APPLICATION OF TRADING COSTS IN THE DECISION-MAKING
PROCESS (TRAINING PERIOD)

Tick	B&H	TC Neglected				TC=0.2%			
		0%	0.1%	0.2%	0.3%	0%	0.1%	0.2%	0.3%
BBDC4	1.982	2.143	2.288	2.288	2.143	1.649	1.882	1.882	1.649
BRKM5	1.902	4.182	4.182	4.182	4.018	4.072	4.072	4.072	3.894
CMIG4	0.573	2.516	2.516	3.586	2.516	1.886	1.886	3.076	1.886
GGBR4	-0.229	1.960	1.466	1.466	1.960	1.840	1.344	1.344	1.840
ITUB4	1.893	0.977	0.926	0.926	1.610	0.429	0.435	0.435	1.064
PETR4	-0.193	2.891	2.891	2.672	2.151	2.759	2.759	2.552	1.987
VALE3	0.828	0.551	0.676	0.579	0.936	0.275	0.251	0.362	0.643
VIVT4	1.739	3.096	3.211	3.497	3.169	2.305	2.444	2.733	2.215

Four simulations were carried out using different values for trading costs, $TC = \{0\%, 0.1\%, 0.2\%, 0.3\%\}$, along with the training period. In the sequence, two experiments were conducted in the validation data: i) no TC were applied to (as detailed in the four columns referred to as “TC Neglected”); ii) a constant TC of 0.2% were considered (as reported in the four columns referred to as “TC=0.2%”). The B&H strategy for each time-series is also presented. The Sharpe Ratio is used as the performance metric.

2) *Break-Even Analysis*: As an alternative to evaluate transaction costs, a “break-even” analysis was carried out. The break-even transaction cost (BEC) is estimated by the ratio of the daily excess returns (beyond B&H) produced by each TS and the number of trades over the entire period [41]. Risk is estimated by the standard deviation of daily returns. Table XII summarizes the key results. Based on the analysis of this table, one can observe that the proposed TS outperforms the CTS in all simulations. Moreover, for BRKM5, GGBR4, and PETR4 the proposed TS surpasses its attendant risks.

VII. DISCUSSION

As stated in [8] “A trader needs to choose between risk and return.” In this sense, an overview of the results indicated that the proposed TS can achieve a similar performance in terms of returns but more favorable in terms of risks.

Although the results in Table VI do not point to a significant profitability of one of the strategies, the results of Table VIII, adjusted by risk metrics, indicate a prevalence of the proposed TS. Comparing the set of TTRs, the proposed TS outperforms the CTS in the majority of the investigated scenarios. This is a strong argument that the use of SSA as a unique framework for detection of long- and short-term movements in stock markets may be rewarding, in line with previous studies [17]–[20].

Regarding the proposed TS, the usage of a pool of TTRs is also worthy. The main idea is that different trading strategies of individual subsystems can offer complimentary information.

TABLE XII
BREAK-EVEN COSTS AND RISK

Method	BBDC4		BRKM5		CMIG4		GGBR4		ITUB4		PETR4		VALE3		VIVT4	
	BEC	Risk	BEC	Risk	BEC	Risk	BEC	Risk	BEC	Risk	BEC	Risk	BEC	Risk	BEC	Risk
SSA	-1.080	1.399	8.246	1.723	1.209	1.772	5.959	1.933	-0.851	1.754	5.020	1.832	-0.415	1.819	0.391	1.244
CTS	-1.818	1.475	-3.027	1.357	0.337	1.330	0.004	1.881	-4.942	1.584	0.103	1.853	-0.629	1.90	-2.440	1.246

BEC is the break-even cost, estimated by the ratio of the daily excess returns (beyond B&H) produced by each TS and the number of trades over the entire period. Risk is the standard deviation of daily returns.

In this spirit, the ensemble could potentially unveil different temporary patterns in the dataset. This process might lead to a trustworthy judgment, since it is expected that the combination of those classifiers provides an unbiased decision. The results presented in Tables VIII and IX confirm this hypothesis.

Another remarkable feature of the proposed ensemble can be seen by the comparison of Tables VIII and IX. In general, the MCDM achieve similar, or even better performances, than a single criterion. For the stock ITUB4, it reveals that the performance by MCDM is superior to the decision with a single criterion when evaluated according to the same criterion. This phenomenon occurs for all metrics, except for Profit Factor.

If a subset of metrics is available, the MCDM using a risk-adjusted ensemble was, in general, more appropriate than the ensemble established using profit-based metrics, as the scenarios presented in Section VI-C reveal. This result corroborates to other studies that deal with a similar topic [36].

There is a tradeoff concerning TC [42]: profit opportunities (which may result in a high number of trades) versus final performance (which may imply in fewer or longer trades). When the TC were considered in the validation process, there were decrements in the performance. The analysis of Table XI does not reveal a regular pattern supporting whether or not the inclusion of TC during the training step is beneficial. Furthermore, regarding the BEC test, for three time-series the proposed TS surpasses its attendant risks. In the same simulations, the CTS does not exceed.

Since only long positions were allowed, it must be highlighted that the TS could not benefit from bearish markets. This constraint arises from the fact that short-position is very limited in the Brazilian stock market, due to: 1) the option market liquidity is limited to PETR4 and VALE3; 2) the participants have to previously rent stocks in order to re-sell them, resulting in higher prices compared to US and U.K. stocks; and 3) instruments such as contract for difference and spread betting are prohibited by regulators. For a recent discussion on this topic, see [43].

VIII. CONCLUSION

It was proposed an adaptive, ex ante TS based on the combination of several SSA-TTRs and the decision-making process inspired in ensembles. Usefulness of this TS system has been illustrated in bullish and bearish markets in ex post simulations, with the performance evaluated by different risk-adjusted and profit-based metrics. Also the effects of transaction costs were assessed. The results suggest that the proposed TS possess strong generalization and adaptability capacities, providing improvements, or at least comparable results, with respect to the B&H and a pool of classic tools, from the realm of TA.

In general, the proposed TS reduced risks in trading, yielding better risk metrics.

The authors are seeking to merge SSA-TTR and AI techniques in future. Finally, a key aspect of this on-going work is to examine SSA-TTR in the derivative markets with less restrictions of short-selling, such as Forex, futures, options, and cryptocurrencies.

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